GROUP ASSIGNMENT

COURSE CODE: PH-101

COURSE TITLE: INTRODUCTION TO LOGIC

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REDUCTION OF ORDINARY LANGUAGE EXPRESSION INTO STANDARD FORM CATEGORICAL PROPOSITIONS

DEFINITION:-

Converting everyday speech into standard form categorical proposition is a process of stripping away the "fluff" of natural language to reveal the underlying logical structure.
A Standard form proposition must follow the structure: [Quantifier] [Subject term] is/are [Copula] [Predicate term].

Meaning of Reduction:

- In daily life, people use long sentences with extra words. These sentences may hide the real logical meaning. Reduction removes the extra wording and expresses the statement in a logical form using standard symbols or simple propositions.

For Example:

- "Most students passed the exam“

Reduction:

- Removes extra words
- Clarifies meaning
- Converts statements into logical form

Reduced form:

- **Some students are people who passed the exam**

Complex Example of Symbolic Form

- **Ordinary Sentence:**
  "If Ali studies and works hard then he will pass or get a job"
- P = Ali studies
- Q = Ali Works hard
- P = Ali Pass
- P = Ali get a job

Symbolic Form :- $(P \wedge Q) \rightarrow (R \vee S)$

Example of Symbolic Reduction

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Reduction:

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Reduced form:

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Four Standard Forms (Basic Types)

- Every categorical proposition must fall into one of these four types:

Type	Name	Form	Example
A	Universal Affirmative	All S are P	All humans are mortal
E	Universal Negative	No S are P	No cats are reptiles
I	Particular Affirmative	Some S are P	Some students are intelligent
O	Particular Negative	Some S are not P	Some birds are not flyers

Structure of Standard Form

- Every standard categorical proposition must follow this structure:
- ☞ **[Quantifier] + [Subject] + [Copula] + [Predicate]**
- Quantifier → All / No / Some
- Subject → the class being discussed
- Copula → is / are / are not
- Predicate → what is being said about the subject

Example:

- All students are intelligent.

Steps to reduce to logical forms:

1. Identify the four standard forms: Every reduction must result in one of these four types:

- **A (Universal Affirmative):** All S are P.
- **E (Universal Negative):** No S are P.
- **I (Particular Affirmative):** Some S are P.
- **O (Particular Negative):** Some S are not P.

Steps to reduce to logical forms:

2. Translation of Quantifiers: Ordinary language often uses non-standard words to indicate quantity. We must translate them into "All", "No", or "Some".

- **Universal (All/No):** Words like every, each, any, anything, whoever, always, and never usually signal a universal statement.

  Example: "Every bird has wings" → **All birds are winged creatures.**

- **Particular (Some):** Words like many, several, a few, most, frequently, and there are signal a particular statement.

  Example: "Most students passed" → **Some students are people who passed.**

Steps to reduce to logical forms:

3. Dealing with Exclusive Propositions: Propositions using "only", "none but", and "none except" are called exclusive. These are tricky because the subject and predicate often switch during translation.

- **The Rule:** "Only P are S" becomes **"All S are P."**

Example: "Only citizens can vote."

- *Reduction:* **All people who can vote are citizens.** (Not "All citizens can vote" which is a different claim).

Steps to reduce to logical forms:

4. Handling Conditional Statements: If-then statements are usually translated as Universal (A or E) propositions.

- **Rule:** If the antecedent (the "if" part) refers to a class of objects, it becomes the Subject.

Example: "If it is a cat, then it is a mammal."

- *Reduction:* **All cats are mammals.**
- **Negative Conditionals:** "If it's not a rose, it's not fragrant."
- *Reduction:* **All fragrant flowers are roses.** (By transposition).

Steps to reduce to logical forms:

- **5. Singular and Pronoun Propositions:** When a sentence refers to a specific person, place, or thing (e.g., "Socrates is mortal" or "This table is heavy"), logic treats it as a universal (A or E) proposition with a class containing one member.
- **Rule:** Use a parameter like "People identical to..." or "Things identical to..."

Example: "Socrates is mortal."

- *Reduction:* **All people identical to Socrates are mortal beings.**

- **6. Converting Adjectives to Nouns:** In categorical logic, the subject and predicate must be nouns or noun phrases (classes), not just adjectives.

Example: "All tigers are carnivorous."

- *Reduction:* **All tigers are carnivores** (or meat-eating animals).

Steps to reduce to logical forms:

7. Exception Statements: Phrases like "All except," "All but," or "Nearly all" are actually compound statements in ordinary language, but for simple categorical reduction, they are often treated as Particular (I or O) because they imply a subset.

Example: "All except seniors are invited."

- *Reduction:* This actually implies two statements: **"No seniors are invited people"** AND **"All non-seniors are invited people."**

RULES FOR REDUCTION

- **Rule 1: Identify Subject (S) and Predicate (P)**
- **Rule:** Every statement has a **subject** (what the sentence is about) and a **predicate** (what is being said about the subject).
- **Example:**
  - **Ordinary sentence:** "Cats sleep."
  - **Subject (S):** Cats
  - **Predicate (P):** Animals that sleep
  - **Standard Form:** All cats are animals that sleep.

- **Rule 2: Add Missing Words**
- **Rule:** Sometimes sentences are incomplete or missing logical words like **"are"** or **"some"**. Add them to form a proper categorical proposition.
- **Example:**
  - **Ordinary sentence:** "Dogs bark"
  - **Missing word added** → "All dogs are animals that bark"
  - **Standard Form:** All dogs are animals that bark.

RULES FOR REDUCTION

- **Rule 3: Change Singular Statements**
- **Rule:** Singular statements about a single person or object are treated as **universal statements**. Use "all" with a class containing only that one member.
- **Example:**
  - **Ordinary sentence:** "Socrates is wise."
  - **Singular → General form:** All people identical to Socrates are wise.
  - **Standard Form:** All people identical to Socrates are wise.

- **Rule 4: Sentences with "Only"**
- **Rule:** When a sentence contains **"only"**, the subject and predicate often switch places to form the correct categorical statement.
- **Example:**
  - **Ordinary sentence:** "Only students can enter the library."
  - **Reduction:** All people who can enter the library are students.
  - **Standard Form:** All people who can enter the library are students.

RULES FOR REDUCTION

- **Rule 5: Sentences with "None"**
- **Rule:** Sentences that use "none" are translated into **universal negative statements (Type E)**.
- **Example:**
  - **Ordinary sentence:** "None of the students were absent."
  - **Reduced form:** No students are absent
  - **Type:** E proposition

- **Rule 6: Sentences with "Not All"**
- **Rule:** Sentences with "not all" are translated into **particular negative statements (Type O)**.
- **Example:**
  - **Ordinary sentence:** "Not all birds can fly."
  - **Reduced form:** Some birds are not able to fly
  - **Type:** O proposition

RULES FOR REDUCTION

- **Rule 7: Statements with "Few"**
- **Rule:** Statements using "few" indicate that only a small portion satisfies the predicate → translated as **particular negative statements**.
- **Example:**
  - **Ordinary sentence:** "Few people are rich."
  - **Reduced form:** Some people are not rich
  - **Type:** O proposition

- **Rule 8: Statements with "Many/Several"**
- **Rule:** Sentences with "many" or "several" indicate **particular affirmative statements (Type I)**.
- **Example:**
  - **Ordinary sentence:** "Many students play cricket."
  - **Reduced form: S**ome students are cricket players
  - **Type:** I proposition

Why is Reduction Important?

- Removes confusion and ambiguity.
- Helps to test whether an argument is valid or invalid.
- Makes reasoning more systematic.
- Used in science, mathematics, computer science, and philosophy.

Benefits of Reduction

- Make reasoning clear and logical
- Helps in Cconstructing **syllogisms**
- Avoids **ambiguity in statements**
- Easy to compare statements in arguments
- Useful in **philosophy, law, and debate**
- Improves **critical thinking skills**
- Example: “All cats are animals” → easy to reason about

Quick Recap

- Identify **subject and predicate**
- Determine **affirmative/negative** & **universal/particular**
- Use **standard forms (A, E, I, O)**
- Replace ordinary words: all, no, some, some…not
- Apply rules: None, Not all, Few, Most
- Examples help **practice reduction**
- **Goal:** Clear & precise logical statements

Conclusion

Conclusion :-

- We have learned how to **reduce ordinary language sentences** into logical forms.
- Identifying the **subject and predicate** is the first step.
- Standard forms **A, E, I, O** make reasoning clear.
- Rules like **None, Not All, Few, Most** help in conversion.
- Practicing examples strengthens understanding and accuracy.
- This process improves **critical thinking and logical reasoning**.
- **Thank you for your attention!**